1(a)	$f(x)$,, 9	B1
		(1)
(b)	$fg(1.5) = f\left(\frac{3}{2\times1.5+1}\right) = 9-\left(\frac{3}{2\times1.5+1}\right)^2$	M1
	$=\frac{135}{16}$	A1
		(2)
(c)	$g(x) = \frac{3}{2x+1} \Rightarrow g^{-1}(x) = \frac{3-x}{2x}$	M1 A1
	$0 < x$,, 3	B1
		(3)
		Total 6

2(a)	$R = \sqrt{5}$	B1
	$\tan\alpha = \frac{2}{1} \Rightarrow \alpha = ...$	M1
	$\alpha = 1.107$	A1
		(3)
(b)(i)	$\text{Max} = 3 + 7\sqrt{5}$	B1ft
(b)(ii)	$\left(2x - "1.107"\right) = \pi \Rightarrow x = ...$	M1
	$\Rightarrow x = \frac{\pi + "1.107"}{2} = 2.12$	A1
		(3)
		Total 6

3(a)	$\log_{10} y = \frac{5}{16}x + 1.5$	M1A1
		(2)
(b)	$\log_{10} y = "\frac{5}{16}"x + "1.5" \Rightarrow y = 10^{"\frac{5}{16}"x + "1.5"}$	M1
	$\Rightarrow y = 10^{"\frac{5}{16}"x} \times 10^{"1.5"}$	M1
	$y = 31.6 \times 2.05^{x}$	A1
		(3)
		Total 5

4(a)	Any correct constant, so for $A=2$ or $B=3$ or $C=-1$ or $D=5$	B1
	$2x^4+15x^3+35x^2+21x-4=Ax^2(x+3)^2+Bx(x+3)^2+C(x+3)^2+D$ $\Rightarrow A=...,B=...,C=...,D=...$ or $2x^4+15x^3+35x^2+21x-4\div(x^2+6x+9)=...x^2+...x+...+\frac{...}{(x+3)^2}$	M1
	2 correct of $A=2,B=3,C=-1,D=5$	A1
	$A=2,B=3,C=-1,D=5$	A1
		(4)
(b)	$\int f(x)\,dx=\int\left(2x^2+3x-1+\frac{5}{(x+3)^2}\right)dx=\frac{2x^3}{3}+\frac{3x^2}{2}-x-\frac{5}{x+3}(+c)$	M1A1ftA1
		(3)
		Total 7

5(a) Way One	$\cot^2 x - \tan^2 x \equiv \dfrac{\cos^2 x}{\sin^2 x} - \dfrac{\sin^2 x}{\cos^2 x} \equiv \dfrac{\cos^4 x - \sin^4 x}{\sin^2 x \cos^2 x}$	M1
	$\equiv \dfrac{\left(\cos^2 x - \sin^2 x\right)\left(\cos^2 x + \sin^2 x\right)}{\sin^2 x \cos^2 x} \equiv \dfrac{\cos 2x}{\ldots}$ or $\dfrac{\ldots}{\left(\dfrac{1}{2}\sin 2x\right)^2}$	dM1
	$\equiv \dfrac{\cos 2x}{\left(\dfrac{1}{2}\sin 2x\right)^2}$	A1
	$\equiv 4\dfrac{\cos 2x}{\sin 2x \sin 2x} \equiv 4\cot 2x \operatorname{cosec} 2x$*	A1*
(b)	$4\cot 2\theta \operatorname{cosec} 2\theta = 2\tan^2\theta \Rightarrow \cot^2\theta - \tan^2\theta = 2\tan^2\theta \Rightarrow \cot^2\theta - 3\tan^2\theta = 0$	M1
	$\cot^2\theta - 3\tan^2\theta = 0 \Rightarrow \dfrac{1}{\tan^2\theta} - 3\tan^2\theta = 0 \Rightarrow \tan^4\theta = \dfrac{1}{3}$	A1
	$\tan^4\theta = "\dfrac{1}{3}" \Rightarrow \tan\theta = \pm\sqrt[4]{"\dfrac{1}{3}"} = "\pm 0.7598.." \Rightarrow \theta = \ldots$	M1
	$\theta =$ awrt $0.65,\ -0.65$	A1A1
		(5)
		Total 9

(a) Way 2	$4\cot 2x\operatorname{cosec}2x \equiv 4\dfrac{\cos 2x}{\sin 2x}\times\dfrac{1}{\sin 2x} \equiv \dfrac{4\left(\cos^2 x-\sin^2 x\right)}{\ldots}$ or $\dfrac{\ldots}{4\sin^2 x\cos^2 x}$	M1
	$\dfrac{\cos^2 x-\sin^2 x}{\sin^2 x\cos^2 x} \equiv \dfrac{1}{\sin^2 x}-\dfrac{1}{\cos^2 x} \equiv \operatorname{cosec}^2 x-\sec^2 x$	dM1A1
	$\equiv 1+\cot^2 x-1-\tan^2 x \equiv \cot^2 x-\tan^2 x$*	A1*

(a) Way 3	$\cot^2 x - \tan^2 x \equiv 4\cot 2x \operatorname{cosec} 2x$ $\dfrac{\cos^2 x}{\sin^2 x} - \dfrac{\sin^2 x}{\cos^2 x} \equiv 4 \times \dfrac{\cos 2x}{\sin 2x} \times \dfrac{1}{\sin 2x}$ $\cos^4 x - \sin^4 x \equiv 4\sin^2 x \cos^2 x \dfrac{\cos 2x}{\sin 2x} \times \dfrac{1}{\sin 2x}$	M1
	$\cos^4 x - \sin^4 x \equiv 4 \times \sin^2 x \cos^2 x \dfrac{\left(\cos^2 x - \sin^2 x\right)}{\left(2\sin x \cos x\right)^2}$	dM1A1
	$\left(\cos^4 x - \sin^4 x\right) \equiv \left(\cos^2 x - \sin^2 x\right)$ $\left(\cos^2 x - \sin^2 x\right)\cancel{\left(\cos^2 x + \sin^2 x\right)} \equiv \left(\cos^2 x - \sin^2 x\right)$ Hence true 1	A1*

6(a)	(i) $P(0,\ 3a)$ (ii) $Q(a,\ 0)$ $R\left(\frac{7}{3}a,\ 0\right)$ (iii) $S\left(\frac{5}{3}a,\ -2a\right)$	B1 B1 B1 B1
		(4)
(b)	$3x-5a-2a=x-2a \Rightarrow x=...$ **or** $-(3x-5a)-2a=-(x-2a) \Rightarrow x=...$	M1
	$x=\frac{5}{2}a$ **or** $x=\frac{1}{2}a$	A1
	$3x-5a-2a=x-2a \Rightarrow x=...$ **and** $-(3x-5a)-2a=-(x-2a) \Rightarrow x=...$	dM1
	$x=\frac{5}{2}a$ **and** $x=\frac{1}{2}a$	A1
		(4)
		Total 8

7(a)	$x = 3\tan\left(y - \frac{\pi}{6}\right) \Rightarrow \frac{\mathrm{d}x}{\mathrm{d}y} = 3\sec^2\left(y - \frac{\pi}{6}\right)$	B1
	$\Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3\sec^2\left(y - \frac{\pi}{6}\right)}$	M1
	$\frac{1}{3\left(1+\tan^2\left(y - \frac{\pi}{6}\right)\right)} = \frac{1}{3\left(1+\left(\frac{x}{3}\right)^2\right)}$	dM1
	$= \frac{3}{x^2+9}$	A1
		(4)
(b)	$y = \frac{\pi}{3} \Rightarrow x = 3\tan\frac{\pi}{6} = \sqrt{3}$	B1
	$x = \sqrt{3} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{3}{\left(\sqrt{3}\right)^2+9}$ or $y = \frac{\pi}{3} \Rightarrow \frac{\mathrm{d}y}{\mathrm{d}x} = \frac{1}{3\sec^2\left(\frac{\pi}{3} - \frac{\pi}{6}\right)} = \ldots$	M1
	$y - \frac{\pi}{3} = \frac{1}{4}\left(x - \sqrt{3}\right)$	dM1
	$y = 0 \Rightarrow 0 - \frac{\pi}{3} = \frac{1}{4}\left(x - \sqrt{3}\right) \Rightarrow x = \ldots$	ddM1
	$x = \sqrt{3} - \frac{4\pi}{3}$	A1
		(5)
		Total 9

8	$\int (2\cos x - \sin x)^2 \, dx = \int \left(4\cos^2 x - 4\sin x \cos x + \sin^2 x\right) dx$	M1
	$\int 4\sin x \cos x \, dx = \int 2\sin 2x \, dx = -\cos 2x$ or $\int 4\sin x \cos x \, dx = -2\cos^2 x \quad \text{or} \quad 2\sin^2 x$	M1
	$\int \left(4\cos^2 x + \sin^2 x\right) dx = \int \left(1 + 3\cos^2 x\right) dx = \int \left(1 + 3\left(\frac{\cos 2x + 1}{2}\right)\right) dx$ or $\int \left(4\cos^2 x + \sin^2 x\right) dx = \int \left(4\left(\frac{\cos 2x + 1}{2}\right) + \frac{1 - \cos 2x}{2}\right) dx$	M1
	$\int (2\cos x - \sin x)^2 \, dx = \frac{3}{4}\sin 2x + \cos 2x + \frac{5}{2}x (+c)$ or $\int (2\cos x - \sin x)^2 \, dx = \frac{3}{4}\sin 2x + 2\cos^2 x + \frac{5}{2}x (+c)$ or $\int (2\cos x - \sin x)^2 \, dx = \frac{3}{4}\sin 2x - 2\sin^2 x + \frac{5}{2}x (+c)$	A1A1
		(5)
		Total 5

8	$\int (2\cos x - \sin x)^2 \, dx = \int 5\cos^2(x + 0.464) \, dx$	M1
	$= \int \frac{5\{\cos(2x + 0.927) + 1\}}{2} \, dx$	M1
	$\frac{5\sin(2x + 0.927)}{4} + \frac{5}{2}x$	M1
	$\frac{3\sin 2x}{4} + \cos 2x + \frac{5}{2}x + c$	A1A1
		(5)
		Total 5

8	$\int (2\cos x - \sin x)^2 \, \mathrm{d}x = \int 5\sin^2(x - 1.107)\,\mathrm{d}x$	M1
	$= \int \frac{5\{1 - \cos(2x - 2.21)\}}{2}\,\mathrm{d}x$	M1
	$= \frac{5}{2}x - \frac{5\sin(2x - 2.21)}{4}$	M1
	$= \frac{5}{2}x + \frac{3\sin 2x}{4} + \cos 2x + c$	A1A1
		(5)
		Total 5

9(a)	$y=\sqrt{3+4e^{x^2}}=\left(3+4e^{x^2}\right)^{\frac{1}{2}}\Rightarrow\frac{dy}{dx}=\frac{1}{2}\left(3+4e^{x^2}\right)^{-\frac{1}{2}}\times 8xe^{x^2}$	M1
	$=4xe^{x^2}\left(3+4e^{x^2}\right)^{-\frac{1}{2}}$	A1
		(2)
(b)	$\frac{\left(3+4e^{x^2}\right)^{\frac{1}{2}}}{x}=4xe^{x^2}\left(3+4e^{x^2}\right)^{-\frac{1}{2}}$	M1
	$\frac{\left(3+4e^{x^2}\right)}{x}=4xe^{x^2}$	dM1
	$4x^2e^{x^2}-4e^{x^2}-3=0*$	A1*
		(3)
(c)	$f(x)=4x^2e^{x^2}-4e^{x^2}-3\Rightarrow f(1)=-3\quad\text{AND}\quad f(2)=652\cdot\ldots$	M1
	Change of sign and f (x) is continuous hence root in (1, 2)	A1
		(2)
(d)	$4x^2e^{x^2}-4e^{x^2}-3=0\Rightarrow x^2=\frac{4e^{x^2}+3}{4e^{x^2}}=\frac{4+3e^{-x^2}}{4}\Rightarrow x=\frac{1}{2}\sqrt{4+3e^{-x^2}}\ *$	B1*
		(1)
(e)(i)	$x_1=1\Rightarrow x_2=\frac{1}{2}\sqrt{4+3e^{-1}}$	M1
	$x_3=1.0997$	A1
(ii)	$\alpha=1.1051$	A1
		(3)
		Total 11

10(a)	$(F=)35$	B1
		(1)
(b)	$200=\dfrac{350\mathrm{e}^{15k}}{9+\mathrm{e}^{15k}}\Rightarrow 1800+200\mathrm{e}^{15k}=350\mathrm{e}^{15k}\Rightarrow 150\mathrm{e}^{15k}=1800$	M1
	$\mathrm{e}^{15k}=\dfrac{1800}{150}\Rightarrow 15k=\ln 12\Rightarrow k=\dfrac{1}{15}\ln 12*$	dM1A1*
		(3)
(c)	$F=\dfrac{350\mathrm{e}^{kt}}{9+\mathrm{e}^{kt}}\Rightarrow \dfrac{\mathrm{d}F}{\mathrm{d}t}=\dfrac{350k\mathrm{e}^{kt}\left(9+\mathrm{e}^{kt}\right)-350\mathrm{e}^{kt}\left(k\mathrm{e}^{kt}\right)}{\left(9+\mathrm{e}^{kt}\right)^2}$	M1A1
	$\dfrac{3150k\mathrm{e}^{kt}}{\left(9+\mathrm{e}^{kt}\right)^2}=10\Rightarrow 315k\mathrm{e}^{kt}=81+18\mathrm{e}^{kt}+\mathrm{e}^{2kt}\Rightarrow \mathrm{e}^{2kt}+\left(18-315k\right)\mathrm{e}^{kt}+81=0$	M1
	$\mathrm{e}^{2kt}+\left(18-315k\right)\mathrm{e}^{kt}+81=0\Rightarrow \mathrm{e}^{kt}=\dfrac{315k-18\pm\sqrt{\left(18-315k\right)^2-4\times 81}}{2}\Rightarrow kt=\ldots$	M1
	$T=$ awrt 5.7, 20.8	A1
		(5)
		Total 9